MBI801: Information Systems Analysis and Design

Assessment Two: Information Systems Analysis and Design

Project Title: Pharmacy management system

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Task 1 CRC Cards Modelling (*by Claire*)

1.1 CRC Cards for Use Case 1- Upload Prescription

Prescription Handler	
• Validate Prescription Format	• Customer
• Validate Prescription Authenticity	• Pharmacist, Doctor
• Encrypt and Store Data	• Database
• Notify Pharmacist	• Notification Service, Pharmacist
• Record Upload details	• Database
• Handle Upload Error	• System

Pharmacist	
• Receive Upload Notification	• Notification Service
• Review Prescription	• Prescription Handler
• Verify Doctor Signature	• Doctor
• Update Prescription Status	• Database
• Notify Customer	• Notification Service

Customer	
• Log In	• System
• Upload Prescription	• Prescription Handler
• Reupload if invalid	• Notification Service
• Receive Reupload Request	• Notification Service
• View Upload Status	• Prescription Handler

Doctor	
• Authorize Prescription	• System
• Confirm Doctor Signature	• Pharmacist

Database	
• Save Prescription Record	• Prescription Handler
• Update Prescription Status	• Pharmacist
• Retrieve Prescription Details	• Pharmacist
• Record Upload Error	• System

Notification Service	
• Send Upload Confirmation • Send Reupload Request • Notify Pharmacist of New Upload • Send Ready for Pickup • Send Stock Alert • Send Error Alert	• Customer, Prescription Handler • Customer, Prescription Handler • Pharmacist, Prescription Handler • Customer • Inventory Manager • Customer, System, Prescription Handler

1.2 CRC Cards for Use Case 2 - Preparing Medication

Pharmacist	
• Log in	• System
• Check Prescription List	• Database
• Review Prescription Details	• Prescription Handler
• Check Stock Availability	• Inventory Manager
• View Medication Detail	• Medication
• Update Prescription Status	• Database
• Notify Customer	• Notification Service, Customer
• Send Hold Status	• Notification Service, Doctor

Inventory Manager	
• Check Stock	• Database, Inventory
• Display Current Stock	• System
• Request Restock	• Supplier
• Notify Pharmacist if Low	• Pharmacist, Notification Service
• Update Stock Quantity	• Database, Inventory
• Retrieve Medication Detail	• Medication

Database	
• Retrieve Prescription Details	• Pharmacist
• Update Prescription Status	• Pharmacist
• Record Preparation	• Pharmacist
• Store Inventory Data	• Inventory Manager

Notification Service	
• Send Ready Notification	• Customer
• Send Stock Alert	• Inventory Manager, Pharmacist
• Send Hold Status Notification	• Pharmacist, Doctor

Customer	
• Receive Pickup Notification	• Notification Service
• Confirm Pickup Delivery	• Notification Service, Pharmacist

Doctor	
• Respond Clarification Request • Verify Prescription Validity • Receive Hold Status Notification	• Pharmacist • System • Notification Service

Inventory	
• Get Stock Level • Adjust Stock Quantity • Update Location • Provide Inventory Data • Retrieve Supplier Information • Store Medication Data	• Database • Database • Inventory Manager • Inventory Manager • Supplier • Medication

Supplier	
• Create Restock Order • Update Delivery Status • Send Confirm Delivery • Provide Supplier Details • Provide Medication Information	• Inventory Manager • Inventory Manager, Database • Inventory Manager, Database • Database • Medication

1.3 CRC Analysis

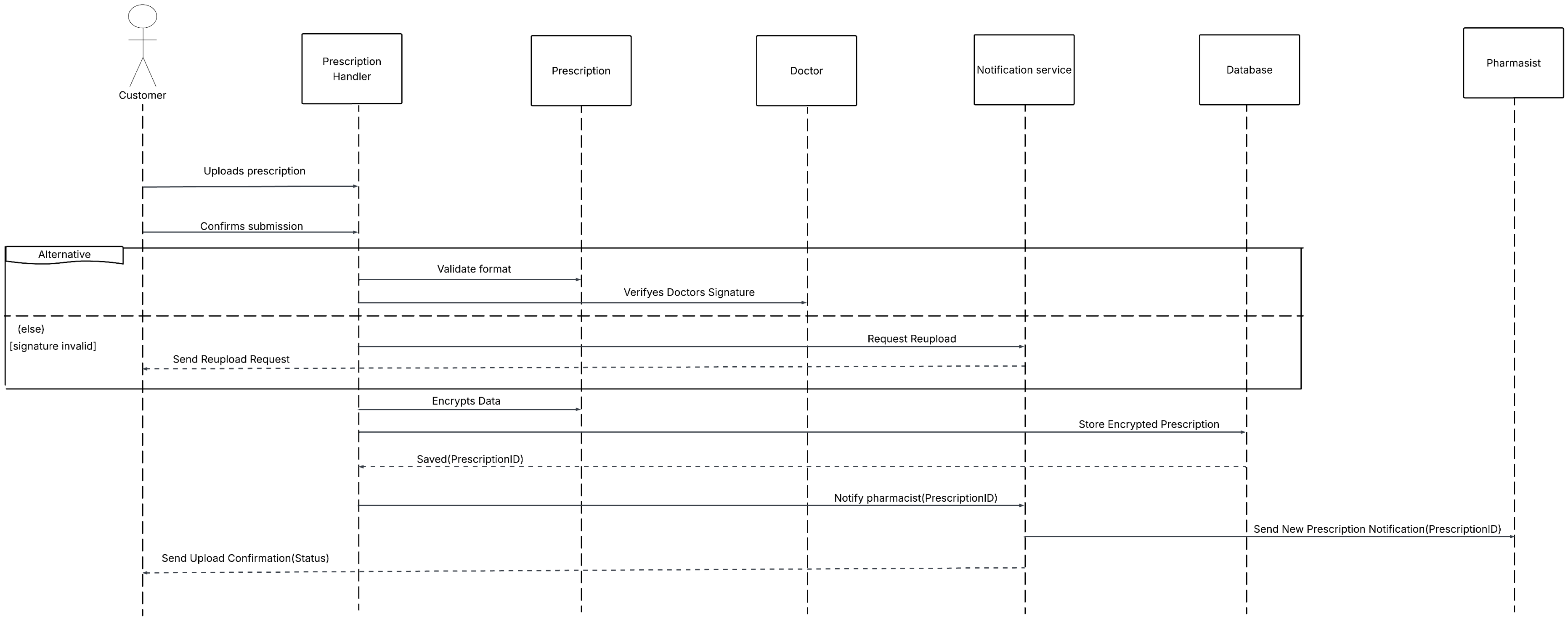
CRC card design helps us to identify the role of each class and their collaboration. It allows us to clearly visualize the division of labour and collaboration among different objects in the system, which provides information for further structured system design diagram such as sequence diagram, class diagram. In the pharmacy management system design, each class is identified based on its business functional role in pharmacy system as well as its position in three layer architecture (view layer, control layer, domain layer).

In the upload prescription process, PrescriptionHandler act as the control class. It's responsible for prescription validation, encryption, and secure storage It collaborates with Customer, Pharmacist, and Doctor to ensure prescription authenticity; interacts with Database for data recording and querying; utilizes NotificationService for automated messaging such as notifying pharmacists of new uploads, confirming successful submissions or requesting uploads again. Therefore, Customer, Pharmacist, Doctor are in view Layer; PrescriptionHandler coordinated workflow processes in controller layer; NotificationService and Database are in domain layer with managing data and process automation.

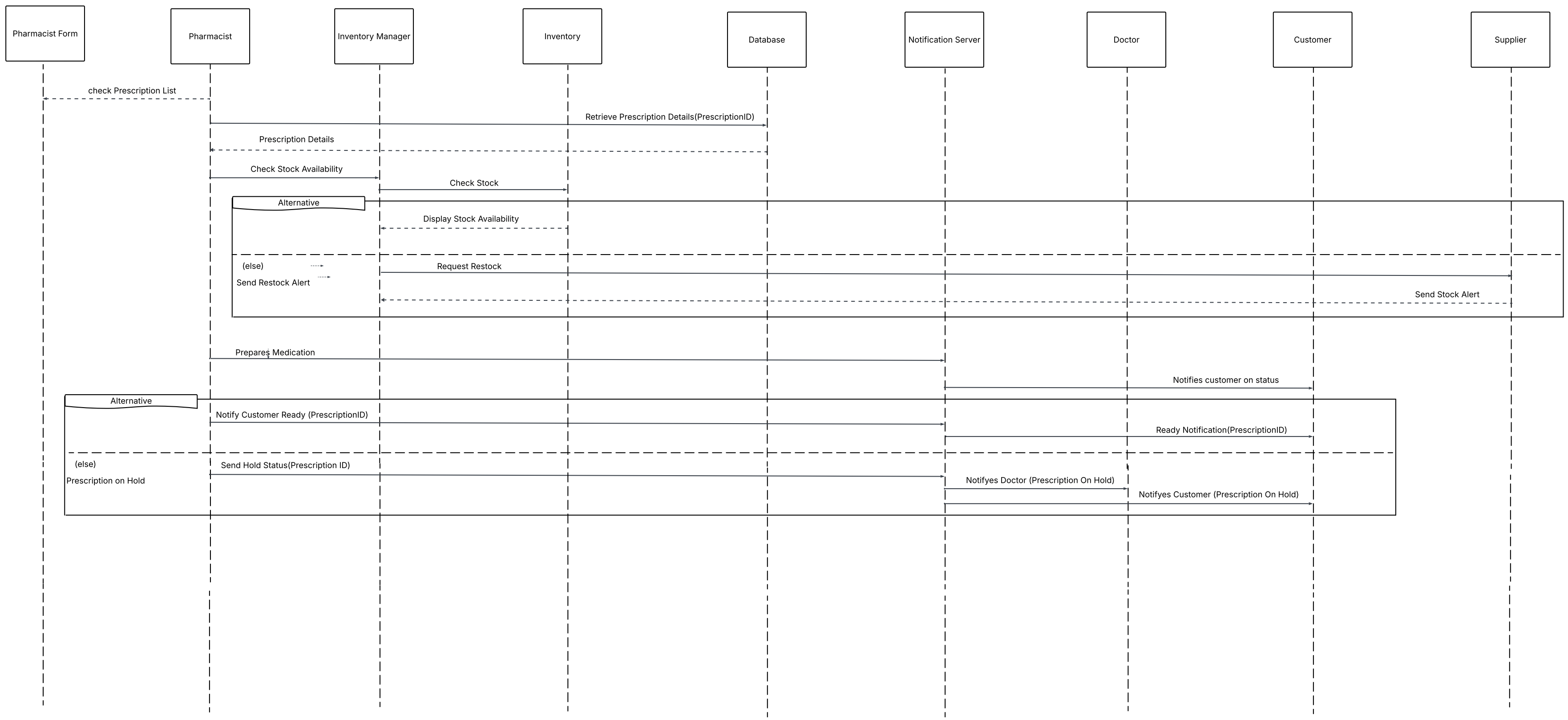
In the preparation medication process, Pharmacist initiate the dispensing, receive prescription approval from Doctor and prepare the medicine. InventoryManager automatic stock verification, update quantity in inventory and ask supplier to replenish. Supplier handle the external supply operation, provide restock information and confirmation to Pharmacist. Database record dispensing process in real time and update prescription and inventory. NotificationService manage the communication between Pharmacist, Customer, send message to Customer medication preparation completed or prescription on hold. This use case also shows three layer architecture: Pharmacist, Customer, Doctor are in view layer; InventoryManager belongs to controller layer; Domain layer include database, Inventory, Supplier, NotificationService.

Task 2 Sequence Diagrams (*by Abirami*)

2.1 Upload Prescription



2.2 Preparing Medication



Task 3 Updated Design Class Diagram (*by Bryan*)

3.1 Use Case 1- Upload Prescription

3.2 Description for Use Case 1

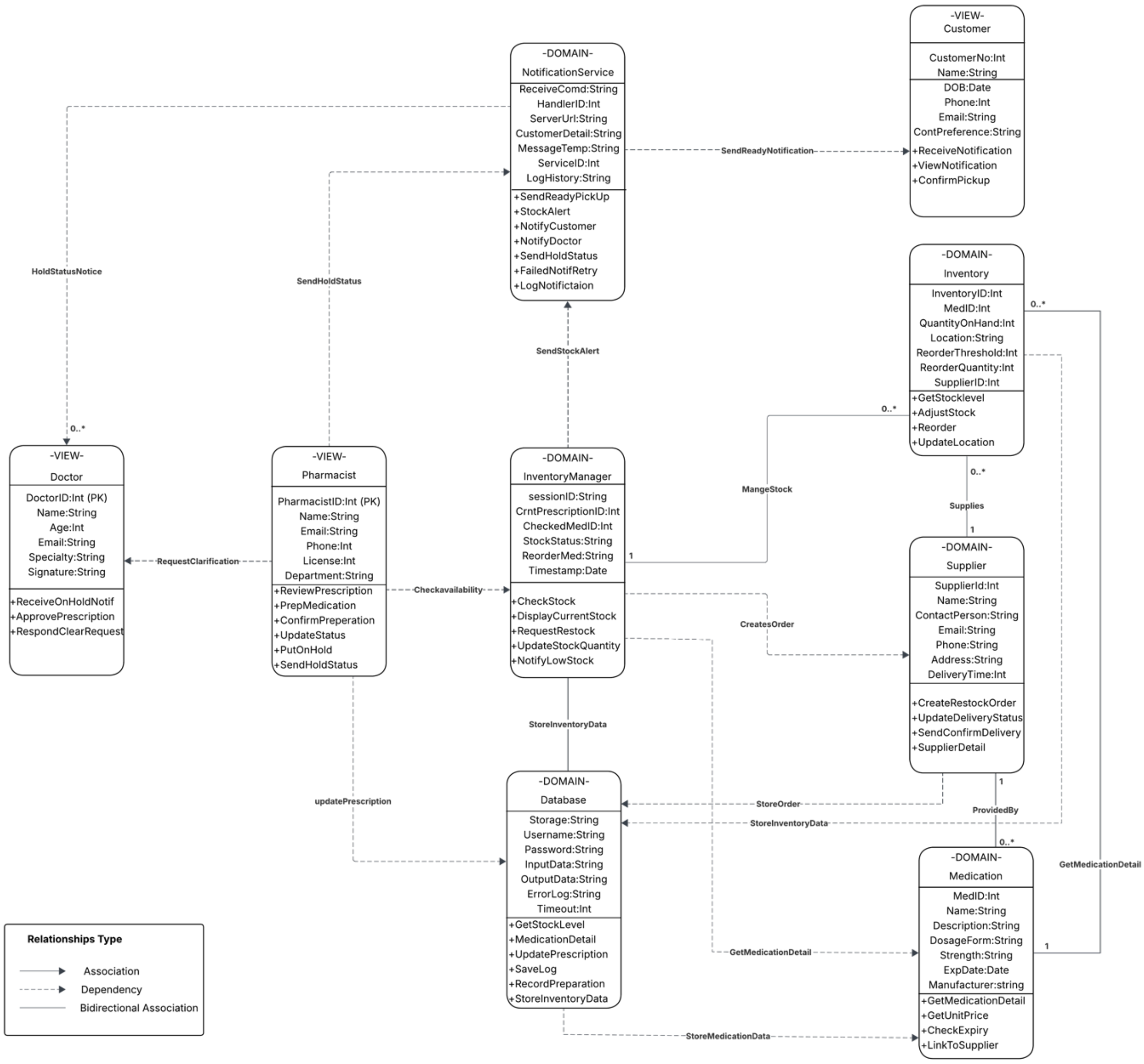
The updated design class diagram for the Upload Prescription use case introduces a clearer separation of responsibilities and stronger data control through a three-layer design. In this model, the Customer, Doctor, and Pharmacist are assigned to the View Layer. These classes are responsible only with the interaction of Upload, verify and check prescription.

The great improvement in this design is the introduction of new class PrescriptionHandler (Controller) which contains the tasks that previously distributed within different classes. It now manages all core operations like validation, encryption, signature verification before interacting with the Database and NotificationService (Domain) classes. The centralization of logic reduces redundancy and better control on process.

The Prescription (Domain) class provides a new data model that stores upload details, dosage, expiry, and status. The Database class stores all the system operations and reduces direct interaction between users and stored information. It acts as central storage, and all the components communicate with this class to reduce duplicate and mislead data in the system. The Inventorymanager monitors stock availability when a new prescription is uploaded. Moreover, it updates the inventory record through Database to ensure that prescribled medicines must be in stock, before approving and preparing. Meanwhile, it sends alert to restock when it is less which can be achieved by NotificationService. NotificationService gives alert to customer and pharmacist through automation. So there is less manual checking and approving, which also reduces the human error and responses to customers quickly.

In conclusion, these enhancements create a more cohesion and efficient flow of data across the system. The View Layer only focuses on user tasks, Controller Layer manages workflow, Domain Layer being data storage and all system core logic. The new design improves the overall maintainability, enhances scalability, and it allows new improvements like automatically verifying the prescription or online Doctor approval without rebuilding the system.

3.3 Use Case 2-Preparing Medication



3.4 Description for Use Case 2

In the class design for the second use case Prepared Medication, on automation and coordination across the three-layer design, on increasing efficiency. In this version, we have interaction between Pharmacist (View), InventoryManager (Controller) and several Domain classes which help us understand how can we make preparing medicine more efficient. Before the preparation starts, the pharmacist got the approved prescription from Doctor and received the notification. Only prescription that has been approved by Doctor can proceed to next preparation process. The pharmacist starts preparation through the interface, but the complex logic, such as verifying stock, updating quantities, or triggering notifications, is now managed entirely within the controller.

The InventoryManager centralizes these actions because it has a connection to the Inventory, Medication, and Supplier classes. The Inventory class holds the stock information, including reorder thresholds and restock quantities. The Medication class holds attributes, including form, strength, and expiry date. he extra Supplier class holds a much more efficient and useful role for re-stocking the medication and updating the supplies when the medication is running low. The InventoryManager also interacts directly with the Database to update the prescription and inverntory records, making sure the records in the system are up to date after each preparation. The Database made sure the record on each medication, prescription, customer, and other classes. It also became a centralised data storage for the whole system. Whereas the NotificationService communicated outcomes, meaning it would notify the customer that their medication was ready for them to pick it up or any other customers of updates, and it would notify the doctor that a prescription was delayed.

By giving each class its own job, the system runs more smoothly and avoids repeating the same data. This makes the system easier to update, and keeps business logic working the same way every time. The new design also leave space for future upgrades, such as real-time tracking of stock and automatically ordering from suppliers.

Task 4 Risk Analysis: System Scalability and Data Integrity (*by Umanga*)

4.1 Scalability Risks

Scalability is the ability of the system to maintain elevating workloads without losing its performance even when the user numbers, prescription numbers, or inventory documents increase. Scalability hazards arise as the pharmacy increases the customer base or incorporates several branches (Rodrigues et al., 2025) . The key risks and corresponding mitigation measures are as follows.

a) **Performance Bottlenecks**

Content: Without optimization of database queries and application logic, system response time may significantly increase alternatively transaction frequence and prescription volumes rise, which adheres to low performance and user dissatisfaction.

Strategy: Conduct regular performance testing to identify and resolve bottleneck actively to ensure consistency.

b) **Inadequate Infrastructure Capacity**

Content: Relying on single on-premise server or ineffectively configured load balancer can cause the system to cause or likely to slow down during high usage period (such as bulk prescription uploads to the system or inventory updates).

Strategy: Implementing appropriate load balancing and caching mechanism should be implemented to balance the workloads, hence maintain high system heavy availability during heave usage.

c) **Ineffective Database Structure**

Content: Without proper indexing and normalization of data, slow data retrieval can occur in the system making the real-time inventory tracking and order processing to be impaired.

Strategy: Adopt a modular architecture of microservices to enable system component independent scaling to improve flexibility and performance.

d) **Integration Limitation**

Content: Since the system is aligned with external platforms such as suppliers, payment gateways, and APIs, the system could face scalability issues if third-party systems fail to handle increased transaction loads.

Strategy: Move to a cloud-based infrastructural set-up (e.g., AWS, Azure) that can support high volume operations and provide auto scaling capabilities.

4.2 Data Integrity

Data integrity provides information accuracy, consistency and reliability across system lifecycle. Since pharmacy information is sensitive, including prescriptions, inventory and patient records, the loss or corruption of data will have a serious impact on operational and compliance (Cai & Zhu, 2015).The key risks and corresponding mitigation measures are as follows.

a) Human Error

Content: Mistakes in manual data entry or inaccuracies during prescription uploads can leading to unreliable information and data.

Strategy: Implement automated data checking and error checking processes when uploading a prescription and order placement to minimise manual mistakes.

b) System Failures or Crashes

Content: Unexpected power failure, database corruption or bugs in software may result in incomplete or lost transactions. Which can lead to privacy breaches and availability issues.

Strategy: Schedule regular database maintenance backups and recovery strategies to prevent data loss.

c) Unauthorized Data Access:

Content: Weak authentication measures can disclose sensitive medical and payment information to unauthorised users. Questioning data privacy and security.

Strategy: Enforce robust authentication and ensure database consistency through the application of ACID principles. (Atomicity, Consistency, Isolation, Duration).

d) Data Synchronization

Content: Data discrepancy could occur if central database isn't synchronised with real time data this usually happens in multiuser operations and in multi branches.

Strategy: Implementation of real time data synchronisation protocols and maintenance of detailed audit trail to monitor and track all changes in inventory management, prescription and payment records could help with unnecessary data loss.

4.3 Conclusion for Two Risks

An effective Pharmacy Management System should focus on high level of scalability and integrity of Data to achieve long term sustainability and uncompromising operations. The system will be able to maintain credible performance and information by using cloud-based infrastructure, data design optimization and stringent data governance practices as the pharmacy network grows. The given strategies help the system to utilise the necessary resources and impact on growth, privacy and security. Moreover, it is effective to reduce the risk of breaching the system, data bias and helps in tracking real time data processing.

Task 5 Cultural and User needs reflection (*by Umanga*)

Guided by cultural awareness and user centred design principle implemented in development of the pharmacy management system to ensure inclusivity, ethical integration and accessibility. Applying cultural perspectives, particularly Te Ao Māori and principles of diversity and inclusion, was central to creating a system that respects Aotearoa New Zealand's bicultural foundation and its multicultural community (Ministry of Business, 2024; Ministry of Health, 2024).

The design process acknowledges the principle from Te Ao Māori perspective, whanaungatanga (relationships), manaakitanga (care and respect), and kaitiakitanga (guardianship). These principles used in the system focus on safeguarding sensitive information about health and prescription data, aligning with the concept of data sovereignty (Ministry of Health, 2020). The systems interface is designed to be culturally responsive, featuring clear and respectful language options, as well as visually simplified layouts are used to accommodate users from diverse cultural backgrounds. Additionally, elements such as easy navigation, readable fonts and bilingual labels ensure accessibility for diverse users, including Māori, Pasifika and elderly patients.

In the user requirement gathering phase, several methods are used such as stakeholder analysis, user goal identification and event decomposition were used to gather insights from customers, suppliers, health inspectors and pharmacists. user personae were created to visualise the different cultural and backgrounds, with behavioural pattern, accessibility preference and digital literacy. Therefore, This approach makes sure that the system acquaint with real world need, like simplifying prescription uploads, enhancing inventory management and reducing waiting times.

User needs were balanced with business efficiency and user satisfaction needs. The core function of the application was around security prescription uploads, automatic inventory updates and real-time order status tracking. These function were ranked highest since it greatly affected operational efficiency as well as patient satisfaction. Meanwhile, Personalised medicine recommendation were planned for later development, which shows the idea of staged implementation approach that considered both resource constraints and user value.

Ethical Consideration and challenges appear when balancing efficiency and cultural sensitivity with users' privacy. For example, ensuring data protection standards require ethical consideration on how patient information, specifically Māori health data is collected and stored. On the other hand, designing for wide set of users should accommodate different levels of digital access and health literacy without being complex.

Overall, this reflection highlights how culturally responsive design and inclusive user engagement informed a system that not only enhances pharmacy operations but also upholds ethical principles, social responsibility, and respect for cultural diversity in New Zealand's healthcare environment.

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Task and Time Assignment

Task Name	Major Leader	Supported Member	Period	Check by others
Task 1:CRC Cards Modelling	Claire	-	3 days	✅
Task 2: Sequence Diagrams	Abirami	-	4 days	✅
Task 3: Updated Design Class Diagram	Bryan	Claire	3 days	✅
Task 4: Risk Analysis (Scalability & Data Integrity)	Umanga	Abirami	Start at anytime- 24 Oct	✅
Task 5: Cultural and User Needs Reflection	Umanga	Abirami	Start at anytime- 24 Oct	✅
Final Document Integrated and Revised	Claire	-	3 days	✅
PPT for Presentation	Claire	Abirami, Umanga, Bryan	3 days	✅
Final Check	All	-	1 days	✅